

Atlantic France Top 50 Content Intelligence

Abstract

This project analyzes the Atlantic France Top 50 music playlist dataset to identify audience preferences, content trends, and ranking characteristics. The analysis focuses on explicit content, album formats, song duration, popularity metrics, and rank-tier behavior.

Introduction

Music streaming platforms generate large amounts of playlist data that can be used to understand listener preferences. This project aims to analyze the France Top 50 playlist and identify patterns in content consumption.

Dataset Description

- Dataset: Atlantic France Top 50 Playlist
- Attributes:
 - Song Name
 - Artist
 - Album Type
 - Popularity
 - Duration
 - Explicit Flag
 - Position
 - Date

Data Preprocessing

- Removed duplicate records
- Removed missing song values
- Converted duration from milliseconds to minutes
- Converted date column to datetime format
- Standardized album type labels
- Created duration buckets
- Created rank-tier categories (Top10, Top25, Top50)

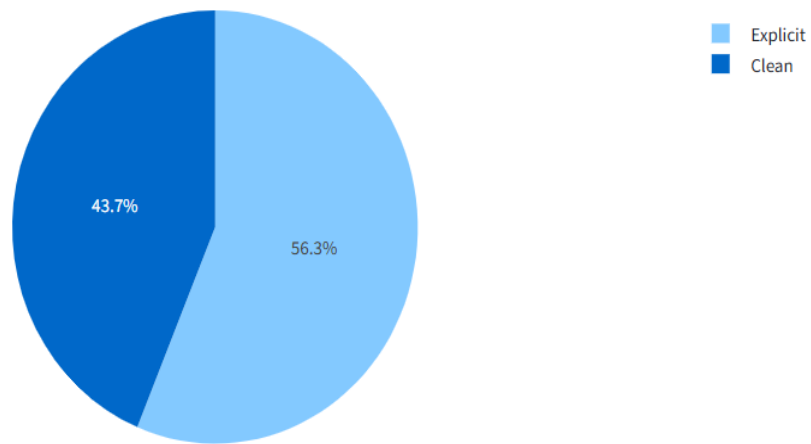
Exploratory Data Analysis

Explicit Content Analysis

- Explicit vs clean content share
- Rank-wise explicit content distribution
- Popularity comparison of explicit and clean songs

Explicit VS Clean Content Analysis

Explicit vs Clean Songs



Interpretation:

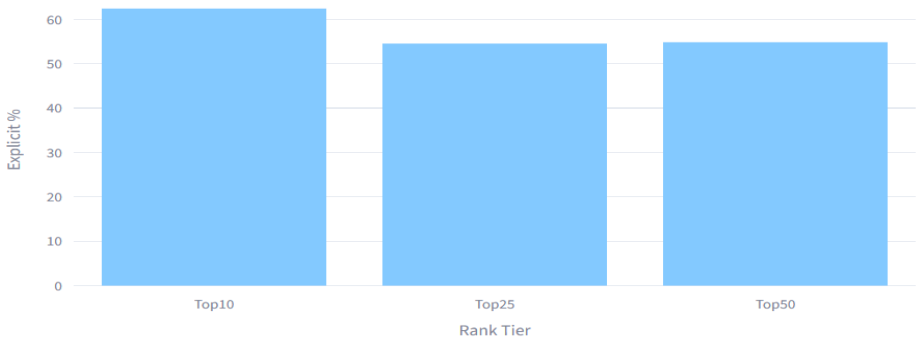
Explains distribution of explicit and clean songs.

Key Insight:

Explains content compliance trend.

Rank Tier Content Comparison

Explicit Content % by Rank Tier



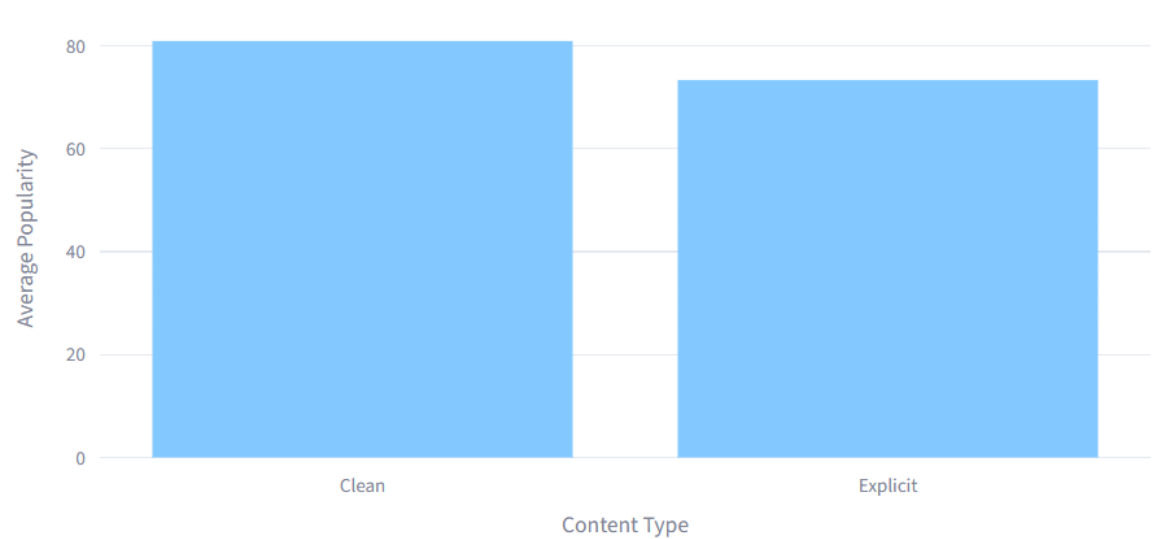
Interpretation:

Shows how explicit content varies across ranking levels.

Key Insight:

Explains whether higher-ranked songs contain more or less explicit content.

Average Popularity: Explicit vs Clean Songs



Interpretation:

Compares average popularity scores of explicit and clean tracks.

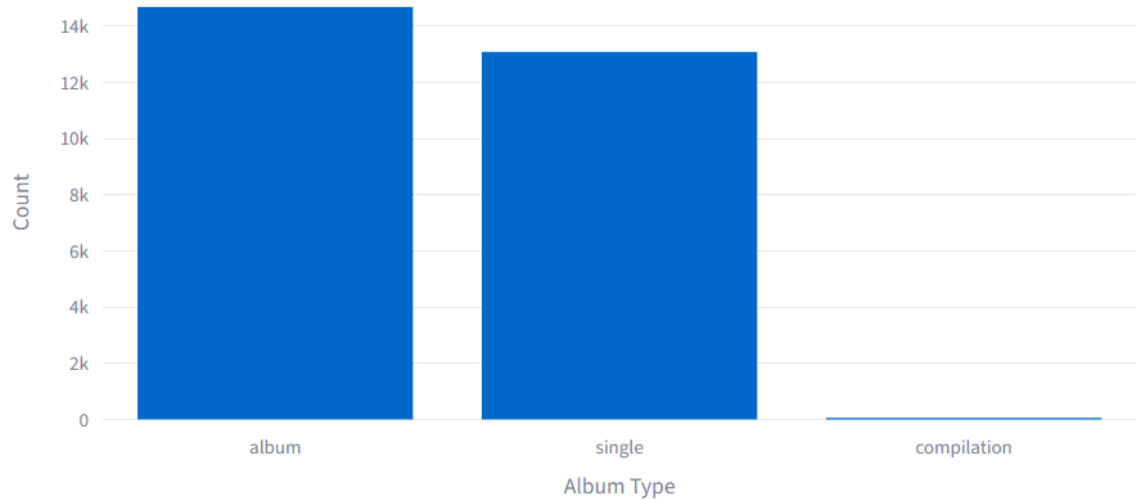
Key Insight:

Shows whether explicit content is associated with higher or lower popularity.

Album Format Analysis

- Single vs album representation

Album Type Distribution



Interpretation

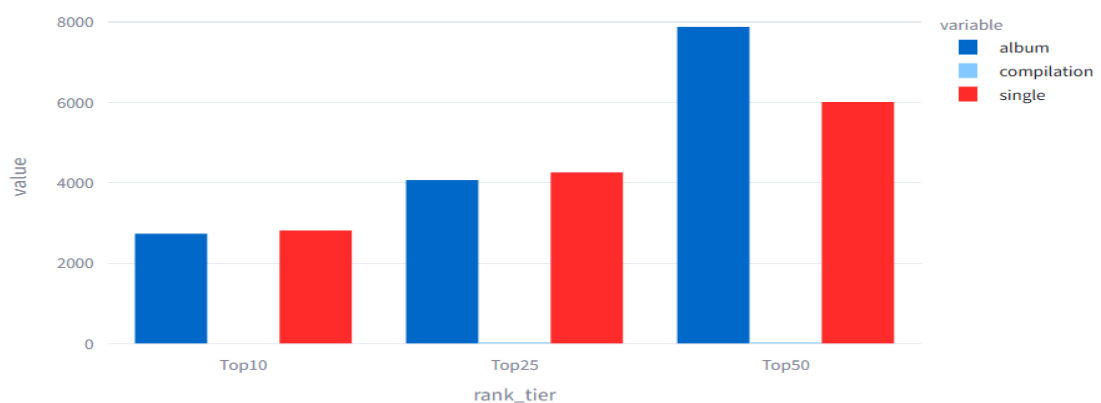
This chart presents the distribution of tracks across different album formats, including singles and albums. The objective is to identify the dominant release format represented in the France Top 50 playlist.

Key Insight

A higher share of singles suggests that standalone releases play a significant role in playlist performance. This reflects modern streaming trends where artists frequently promote individual tracks rather than complete albums.

- Album type popularity comparison

Album Type Distribution by Rank Tier



Interpretation

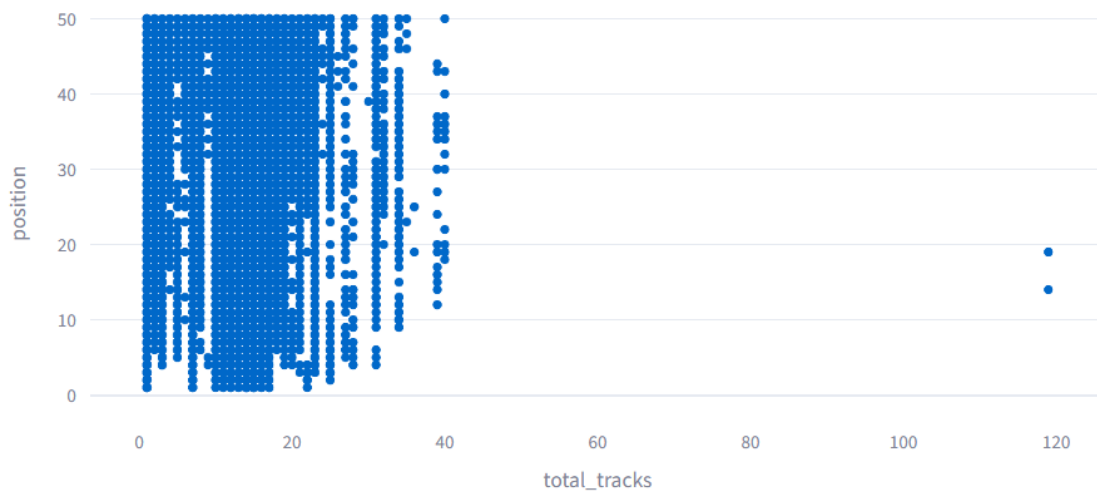
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Key Insight

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- Rank-tier album distribution

Album Size vs Ranking Position



Interpretation

This chart examines how album formats are distributed across Top 10, Top 25, and Top 50 ranking categories. The analysis helps identify whether specific release formats perform better within higher-ranked positions.

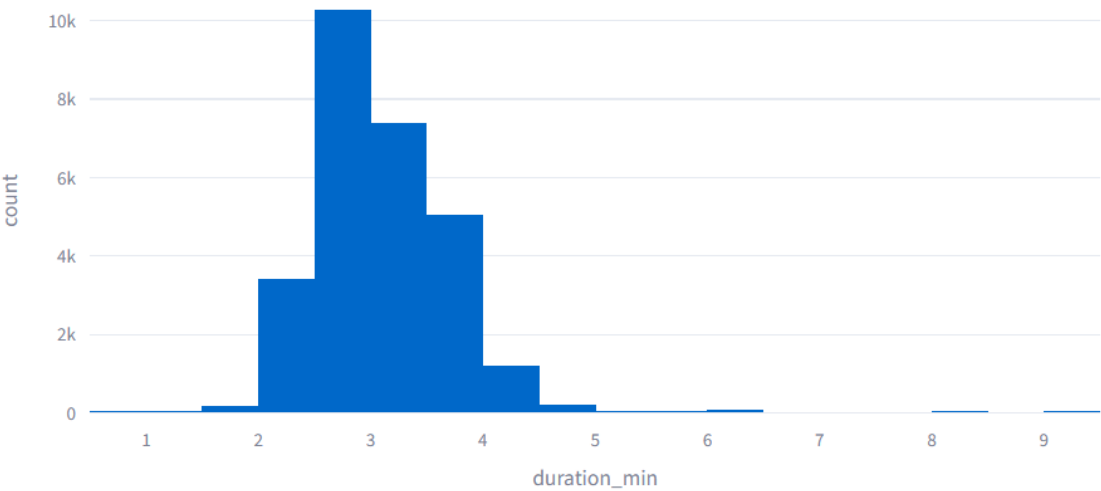
Key Insight

The distribution highlights the relationship between release strategy and ranking success. Greater representation of a particular format within higher ranks may indicate stronger audience preference for that release type.

Duration Analysis

- Duration histogram

Song Duration Histogram



Interpretation

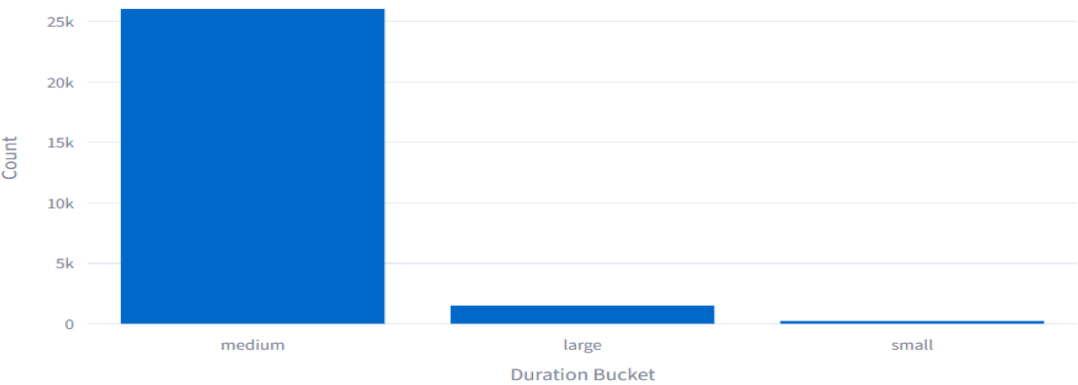
The histogram displays the distribution of song durations across the playlist. Duration analysis helps identify common track lengths and listener consumption patterns.

Key Insight

Concentration of songs within a specific duration range suggests a preferred song length that aligns with audience engagement and streaming platform dynamics.

- Duration bucket analysis

Duration Bucket Distribution



Interpretation

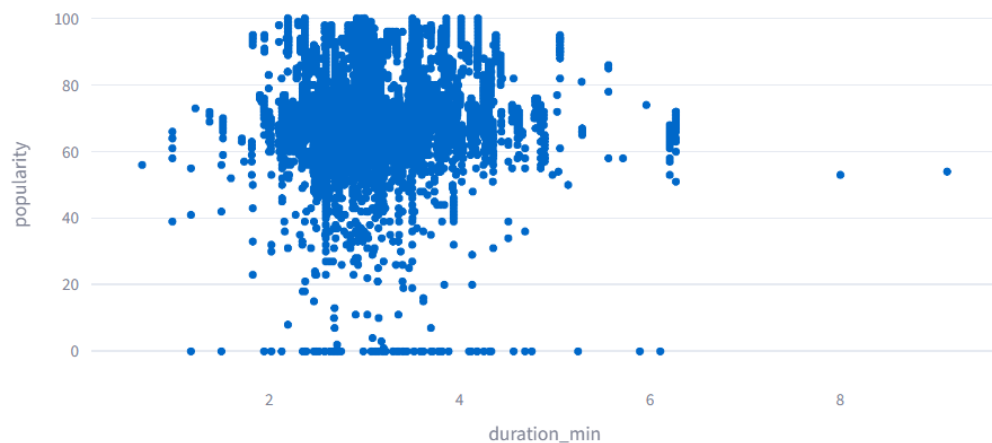
This chart categorizes songs into predefined duration groups such as short, medium, and long tracks. The purpose is to identify the duration profiles most commonly represented within the France Top 50 playlist.

Key Insight

The dominant duration bucket reflects listener preferences and streaming platform trends. A higher concentration within medium-duration tracks may indicate an optimal balance between listener retention and content consumption behavior.

- Duration versus popularity trends

Duration vs Popularity



Interpretation

This scatter plot explores the relationship between song duration and popularity. Each point represents a song, allowing assessment of whether track length influences audience engagement and streaming performance.

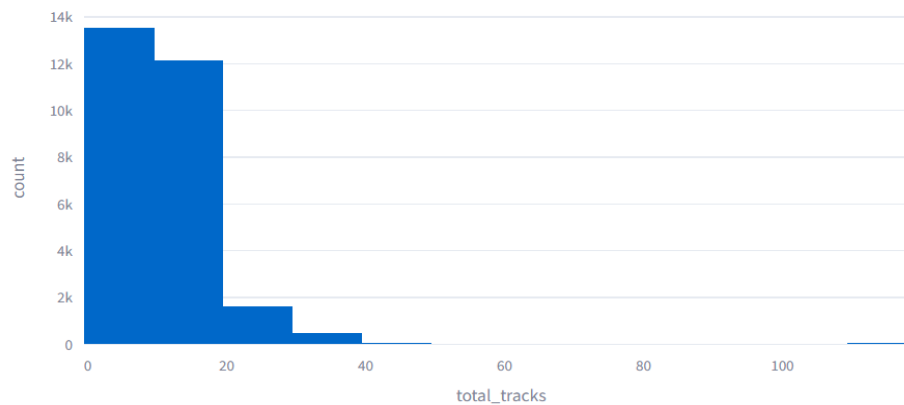
Key Insight

The distribution of points indicates whether song duration has a meaningful impact on popularity. A weak relationship suggests that factors other than duration contribute more significantly to audience engagement.

Album Structure Analysis

- Album size distribution

Album Size Distribution



Interpretation

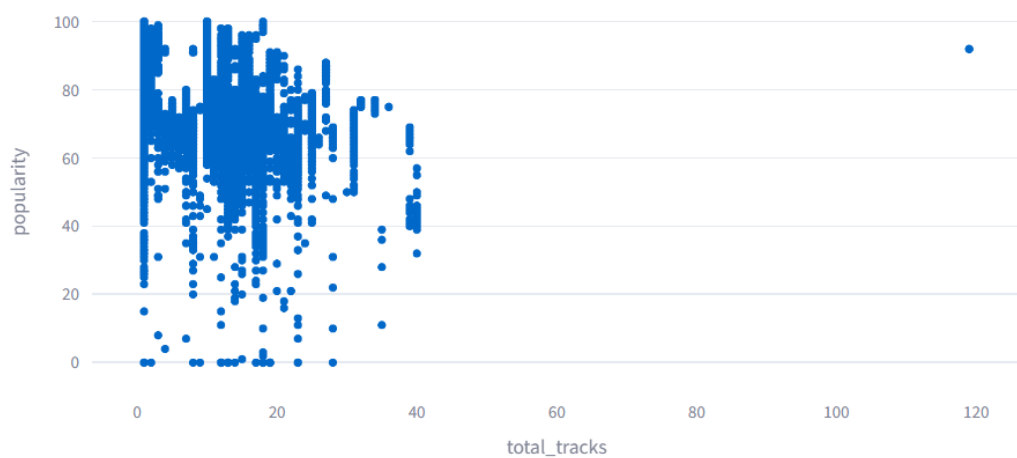
This histogram illustrates the distribution of album sizes based on the total number of tracks available in each album. The analysis helps understand whether songs appearing in the France Top 50 playlist are primarily associated with smaller releases, medium-sized albums, or larger music collections.

Key Insight

The distribution reveals the most common album sizes represented in the playlist. A concentration of tracks from smaller albums may indicate a preference for focused releases, while broader representation from larger albums suggests stronger catalog visibility.

- Album size versus popularity

Album Size vs Popularity



Interpretation

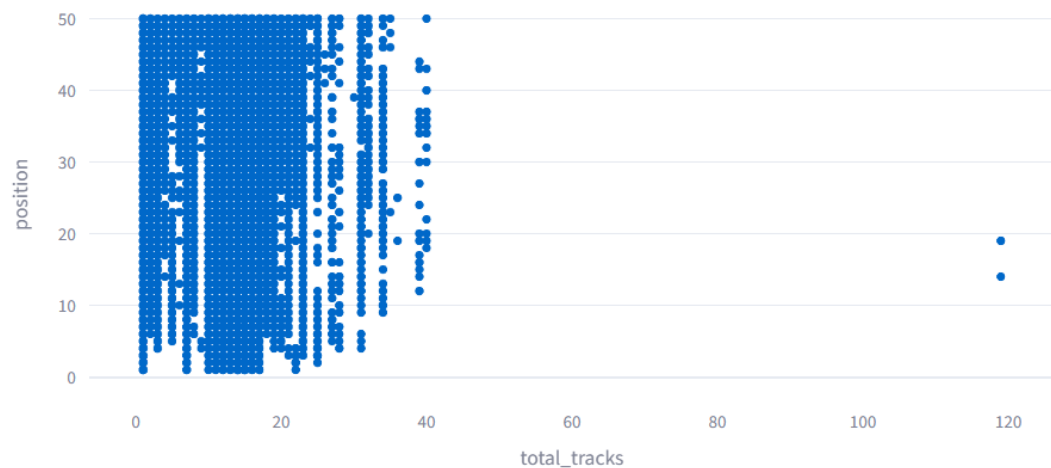
This scatter plot examines the relationship between album size and song popularity. Each point represents a track, with album size measured through total tracks and popularity indicating audience engagement.

Key Insight

The relationship between album size and popularity appears limited when data points are widely dispersed. This suggests that factors such as song quality, promotion, and audience preferences may influence popularity more strongly than album size alone.

- Album size impact on ranking

Album Size vs Ranking Position



Interpretation

This chart evaluates whether album size influences playlist ranking positions. The analysis compares the number of tracks within an album against the song's position in the France Top 50 playlist.

Key Insight

The results indicate whether tracks from larger albums consistently achieve higher rankings or whether ranking success is independent of album size. A weak relationship suggests that individual track performance is more important than the overall size of the album.

Key Performance Indicators

- Explicit Content Share

Interpretation : Measures the percentage of songs classified as explicit within the playlist. This KPI serves as an audience sensitivity indicator and helps assess overall content composition.

- Clean Content Dominance Ratio

Interpretation : Compares the number of clean songs to explicit songs. A higher ratio indicates stronger content compliance and greater representation of clean content.

- Single vs Album Track Ratio

Interpretation : Measures the distribution of tracks across release formats and provides insight into content publishing strategies within the playlist.

- Average Song Duration

Interpretation : Represents the average track length across the playlist and helps identify structural norms in music consumption behavior.

- Album Size Impact Index

Interpretation : Evaluates the relationship between album size and playlist performance, providing insight into catalog strategy effectiveness.

- Content Acceptance Score

Interpretation : Measures the degree to which content characteristics align with ranking performance and audience preferences.

Dashboard Development

A Streamlit dashboard was developed with:

- Date range filters
- Rank tier filters
- Album type filters
- Explicit content filters
- Interactive charts and KPI panels

Dashboard Filters

Select Date Range

2024/05/18 – 2025/11/27

Rank Tier

Top10 × Top25 ×

Top50 ×

Album Type

single × album ×

compilation ×

Explicit content

☒ ALL

☐ Explicit

☐ Clean

Interpretation:

A Streamlit dashboard was developed to provide interactive exploration of the dataset. Users can apply filters based on date range, rank tier, album type, and explicit content while viewing real-time analytics and KPI metrics.

Findings

- France playlists show measurable differences between explicit and clean content.
- Singles dominate playlist representation.
- Song duration tends to cluster around common streaming-friendly lengths.
- Rank tiers reveal differences in content concentration and popularity.

Recommendations

1. Content creators should maintain a balanced mix of explicit and clean content to appeal to diverse audience segments.

2. Artists may consider releasing standalone singles strategically, as singles demonstrate strong representation within playlist rankings.
3. Songs should be optimized around commonly preferred duration ranges to maximize listener engagement and retention.
4. Album planning should prioritize track quality over album size, as larger albums do not necessarily guarantee stronger popularity or ranking performance.
5. Continuous monitoring of content trends through analytics dashboards can support data-driven decision-making for music platforms and content stakeholders.

Conclusion

The analysis provides insights into music consumption trends in France and demonstrates how data analytics can be used to understand playlist composition and audience preferences.

Future Improvements

- Sentiment analysis of lyrics
- Artist-level trend analysis
- Cross-country playlist comparison
- Predictive ranking models